

SDS (SAFETY DATA SHEET)

8 gram Compressed Gas (N2O)filled charger

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND THE COMPANY

COMPANY:GUANGDONG HUATE GAS CO.,LTD

CONTACT:Jinfeng Road,Heshun Lishui Town,Foshan,China TEL:86-757-85123408 FAX:86-757-85123408
COMPANY:ASIA PACIFIC GAS ENTERPRISE COMPANY LIMITED. (HK Branch)

CONTACT:Flat C, 2/F,Block 10,Locwood Court,Kingswood Villas,Tin Shui Wai,N. T.,Hong kong
TEL:(852)8107385 FAX:(852)81073852

PRODUCT:Nitrous oxide supplied in cartridges

FORMULA:N2O

EMERGENCY CONTACT:86-757-81008535

COMPOSITION/INFORMATION ON INGREDIENTS

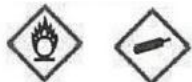
Ingredients	CAS NO	EINECS
Nitrous oxide	10024-97-2	233-032-0
2. HAZARD IDENTIFICATION		

Small gas cartridges without a release device,non refillable with a capacity not exceeding 50ml(Nitrous Oxide), containing only non-toxic constituents are not subject to IMDG Code as per S.P.191(ADR/IMO)&A98(IATA).

2.1. Classification according to Regulation No 1272/2008/EC

GHS label elements

GHS03 GHS04



Category

Oxide,gas:1 category,Gases under pressure,liquefied gas

Hazard Statements phrases)

H270-May cause or intensify fire;oxidiser.

H280-Contains gas under pressure;may explode if heated.

Precautionary Statements(P-phrases)

P244-Keep reduction valves free from grease and oil.

P220-Keep/Store away from clothing/.../combustible materials.

P370+P376-In case of fire:Stop leak if safe to do so.

P403-Store in a well-ventilated place.

Supplemental hazard information



2.2.Classification according to Directive67/548/EEC or Directive 1999/45/EC

O-oxidizing

Xn-harmful



Date:2017-5-17

Date of Revision:2024-11-14

Nitrous oxide supplied in cartridges

Site2/9

Version:7

Risk phrases

R8-Contact with combustible material may cause fire.

R40-Limited evidence of a carcinogenic effect.

Danger signs

S1/2-Keep locked up and out of the reach of children.

S9-Keep container in a well-ventilated place.

S17-Keep away from combustible material

S21-When using do not smoke.

S36/37/39-Wear suitable protective clothing and eye/face protection.

S38-In case of insufficient ventilation,wear suitable respiratory equipment

S45-In case of accident or if you feel unwell,seek medical advice immediately (show the label where possible).

2.3. Other dangers**2.3.1.Healtheffects**

risk of suffocation,

2.3.2.Environmental effects

no information available,

2.3.3.Physicaland chemical danger

oxidizing agent,assisted combustion,

2.3.4. Specific no data**3.COMPOSITION/INFORMATION ON INGREDIENTS**

10024-97-2 Nitrous oxide supplied in cylinder

min.99%



8-40

GHS03 GHS04



Oxide gas:1 category,Gases under pressure,liquid gas,H270-H280

(You can see full text of R and H sentences at 16 point.)

**4.FIRST AID MEASURES****Inhaling**

remove the wounded to fresh air,when unconscious without delay artificial respiration or oxygen inhalation medical treatment is needed.

Eyes

wash out immediately with plenty of water for at least 15minutes.

Skin

risk of chilblain,thaw frostbite slowly with lukewarm water,prohibited to remove clothes,medical treatmentis needed.

Swallowed

no adverse effects anticipated,Do not eat or drink during work-no smoking.

Symptoms and effects,both acute and delayed

inhalation of small amounts cause laughter,intoxicating,pulsating sound and visual hallucinations,ingreater amount cause anesthetic and narcotic (laughing gas)

Date:2017-5-17

Date of Revision:2024-11-14

Nitrous oxide supplied in cartridges

Site:3/9

Version:7

General



Instructions for the doctor:symptomatic treatment,

5.FIREFIGHTING

MEASURES

Fire-hazard class:C Highly flammable



5.1. Adequate extinguishing materials:

use extinguishing material suitable to the near fire circumstances,

5.2. Inadequate extinguishing

materials water jet prohibited,

5.3. Dangerous decomposition

products no data

5.4. Special protection equipment for firemen

Wear self-contained breathing apparatus pneumatic breathing-mask,closed industrial protecting clothing,

5.5. Specific dangers

no data

5.6. Specific methods

no data

6.ACCIDENTAL

RELEASE

MEASURES

6.1. Measures for personal protection:

put on the protective equipment,evacuate the area and close for the traffic,keep upwind,

6.2. Measures for environment protection

if possible stop the leakage,prevent that the substance in gutter,hole and collar gets,when the substance gets in gutter or water,notify the local authorities,

6.3. Methods of decontamination

let's fresh air in the area,warn everybody,when in liquid form is:do not use sawdust or other combustible absorbents, enclose with dam the out streamed substance,cover blot off with dry sand,suitable material,in closed vessel carry to destroying place ,

6.4.Reference to other sections

Find the personal protective equipment's in the 8.

7.HANDLING

AND

STORAGE

7.1. Handling

7.1.1.Technical precautions

ventilation should be provided,it is recommended to use protective equipment's,

7.1.2.Safe handling guidance

avoid the product getting on the skin,in the eyes,on the clothes,avoid breathing itsfume,prevent that the substance comesincontactwith water,equipmentthatissuitable forthisproductandforthe designedpressureandtemperature should be used.

7.2. Storage

7.2.1.Technical measures/storage conditions

in a cool,dry,well ventilated place,Store in a cool,dry well-ventilated area.keep away from heat,sources of ignition,in original,closed packing,protected direct sunlight,



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Date:2017-5-17	Nitrous oxide supplied in cartridges	Site:4/9
Date of Revision:2024-11-14		Version:7

- 7.2.2.Incompatible products
reacts violently with Sulphurous anhydride,amorphous boron,phosphine,ethers,aluminum,hydrazine,phenyl lithium and tungsten carbide causing fireand explosion hazard,the gas strong oxidising agent above 300°C and may form explosive mixtures with ammonia,carbon monoxide,hydrogen sulfide,oil,grease and fuels,
- 7.2.3.Packaging materials
in retail boxes and/or inbulk
- 7.3. Special application
no data

8.EXPOSURE	CONTROLS-PERSONAL	PROTECTION
8.1. Exposure limit values ingredients: 1.Nitrous oxide supplied in cylinder	AKvalue(mg/m3) 180	ckvalue2m0gm ³) MK value(mg/m3)
8.2. Exposure control ventilation should be provided,it is recommended to use protective equipment,avoid breathing its fume, avoid the product getting on the skin,in the eyes,on the clothes,		
8.2.1.Workplace exposure control		
8.2.1.1.Respiratory protection:  inadequate exhaust or ventilation use respiratory protection,in enclosed space a respirator that is independent from the ambient air should be used,Use self-contained breathing apparatus.		
8.2.1.2.Hand protection:  protective gloves,		
8.2.1.3.Eyes protection:  tightly closing protective goggles(EN 166)		
8.2.1.4.Protection of skin and body surface: protective apron,safety shoes(MSZ EN 345-1,SB)		
8.2.2.Environmental exposure control prevent thatthe substance ingutter,hole and collar gets,		

9.PHYSICAL AND CHEMICAL PROPERTIES
9.1. General information Appearance: liquefied gas, Colour: colourless, Odour: sweetish odor, Information on basic physical and chemicalproperties Ph: nincsadat/no data Boiling Point(°C): -88,5°C Ignition temperature(°C): nincs adat/nodata Explosive limit: -lower: nincsadat/nodata -upper: nincsadat/no data



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Date:2017-5-17	Nitrous oxide supplied in cartridges	Site:5/9
Date of Revision:2024-11-14		Version:7
Oxidize facility: nincs adat/no data Vapor Pressure (20° C): nincs adat/no data Gravity (20C): nincs adat/no data Specific gravity(water=1), (20° °C): nincs adat/no data Solubility: -Water: 2,2 mg/ -Other solvents: no data Distribution coefficient(n-octanol/water): nincs adat/no Viscosity: data nincs Vapor Density: adat/no data 1,5 Evaporation Rate: nincs adat/no data Decomposition Temperature: nincs adat/no data Other items: critical temperature:36.4℃,critical pressure:72.4 bar,the critical density:0.452 g/cm3,Filling pressure:7MPa(1,015psi)Pressure at 50℃:16,5MPa(2,425psi)Burst pressure:min.45 MPa		
9.2.	Other Information	
	Melting Point(°C):	-90,81℃
	Flash-point (C):	nincs adat/no data
10. STABILITY AND REACTIVITY		
10.1.Reactivity stable under normal conditions, 10.2.Chemical stability there is no decomposition under normal condition, 10.3.Possibility of hazardous reactions reacts violently with sulphurous anhydride,amorphous boron,phosphine,ethers,aluminum,hydrazine,phenyl lithium and tungsten carbide causing fire and explosion hazard, 10.4.Conditions to avoid keep away from heat,flames,sparks and ignition sources, 10.5.Incompatible materials organic,combustible material, 10.6.Hazardous decomposition products nitrogen monoxide		
11. TOXICOLOGICAL INFORMATION		
Actual material Nitrous oxide supplied in cartridges 11.1.Acute toxicity LD50:no data LC50:1068mg/m3(4 hours) 11.1.1.After inhaling In animals: LC50:1068mg/m3(4 hours)rats inhalation. 11.1.2.When swallowed In animals: no data 11.1.3.In case of contacting skin In animals: no data		



Date: 2017-5-17

Date of Revision: 2024-11-14

Nitrous oxide supplied in cartridges

Site: 6/9

Version: 7

11.2. Local effects

11.2.1. After inhaling

no data

11.2.2. In case of contacting

skin no data

11.2.3. In case contacting

eyes no data

11.3. Sensibility

no data

11.4. Chronic

toxicity no data

11.5. Specific effects

11.5.1. Genotoxicity

no data

11.5.2. Carcinogenicity

no data

11.5.3. Reproductive

toxicity no data

12. ECOLOGICAL INFORMATION

Actual material

Nitrous oxide supplied in cartridges

12.1. Ecotoxicity

12.1.1. Water toxicity

no data

12.1.2. Terrestrial

toxicity no data

12.1.3. Behaviour in waste water treatment

plants no data

12.2. Mobility

no data

12.3. Biological persistence

12.3.1. General

no information available

12.3.2. In water

no data

12.3.3. In air

no data

12.3.4. In soils and

sediments no data

12.4. Bioaccumulative

potential no data

12.5. Results of PBT and vPvB

assessment no data

12.6. Other adverse effects

prevent that the substance in water, agricultural land and gutter gets, when the substance gets in gutter or water, notify the local authorities,



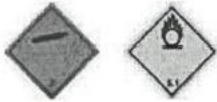
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Date:2017-5-17	Nitrous oxide supplied in cartridges	Site:7/9
Date of Revision:2024-11-14		Version:7

13.DISPOSAL CONSIDERATIONS

- 13.1.Material disposal
- Any product originated from the use of the substance or preparation and unsuitable for further utilisation,as well as the tools and packing materi-als,should be delivered to an approved deposit for recy-cling or neutralisation. observe all state and local environmental regulations,
- 13.2.Package disposal
- observe all state and local environmental regulations,the packaging has to be used,like product
- 13.3. EWC number
- 150105

14.TRANSPORT INFORMATION

- UN number
- 2037
- ADR/ADN/RID
- Proper shipping name: Receptacles,small,containing gas(Gas Cartridges)
- Technical/Trade name: Nitrous oxide supplied in cartridges
- Hazard class: 2
- Labels: 2.2+5.1
- Classification code: 50
- Packing group:
- Hazard identification number: 25
- Hazard of environment: NO
- Tunnel restriction code: (E)



- IMDG
- Propershipping name: Receptacles,small,containing gas(Gas Cartridges) without a release device,non-refillable
- Special provisions: SP191
- Technical/Trade name: Nitrous oxide supplied in
- name:Hazard class: cartridges 22.2+5.1
- Labels:
- Packing F-D,S-U,
- group:EmS: NO

- IATA
- Marine pollutant:
- Proper shipping name: Gas cartridges
- Technical/Trade name: Nitrous oxide supplied in cartridges
- Hazard class: 2.2
- Labels: 2.2(5.1)
- Packing group: -
- PAX: 203
- CAO: 203
- UN Number: 2037



15.REGULATORY INFORMATION

- 15.1.Seveso category
- 3.-Oxidising agents,
- 15.2.Storage category
- 2A storage class:Compressed,lique ed or dissolved under pressure gases,

Date:2017-5-17

Date of Revision:2024-11-14

Nitrous oxide supplied in cartridges

Site:8/9

Version:7

15.3. WGK
WGK 1**15.4.Important national regulations related to the product**

Act No.XCII/1993 on work safety,and the related regulations
Act No.XLII/2000 on waste management
Act No.XXV/2000 on chemical safety
Gov.Decree No.98/2001(VI.15.)on the conditions of pursuing waste-related activities
Decree No.25/1996(VII.28.)of the Minister of Public Health on the general health requirements of working and work conditions without health risks
Decree No.26/1996(VII.28.)of the Minister of Public Health on the limitations of the(weekly,daily) exposure time of workers employed under certain health risks
Decree No.28/2011.(IX.6.)of the Minister of Internal Affairs on issuing the National Fire Protection Rules
Decree No.44/2000.(XII.20.)of the Minister of Health on the detailed rules of certain procedures and activities related to hazardous substances and hazardous products
Joint Decree No.25/2000(IX.30.)of the Minister of Health and of the Minister of Social and Family Affairs on the chemical safety of workplaces,
EUM decree no.
The European Community and Regulation 1907/2006/EC on the registration,evaluation,approval and restriction of chemicals (REACH)

15.5.Other relevant national regulations

Joint Decree No.25/2000(X.30.)of the Minister of Health and of the Minister of Social and Family Affairs on the chemical safety of workplaces
EWC code numbers:Decree No.16/2001(VII.18.)of the Minister for the Environment on the lists of waste,as amended by Decree No.10/2002(I.26.)of the Minister for the Environment
Road Transport Department:Act 19 of 1979 and decree no.712011(X1.8.)amending KPM decree no.20/1979(IX 18.)KPM on the promulgation of Annex"A"and"B"of the European Agreement on the International Road Transport of Hazardous Goods,and Annex C of Act LXXVII of 2006(GKM decree no.72/2011(X1.8.)NFM on the promulgation of the domestic use of the annexes of the International Railway Goods Transport Convention
1272/2008/EC regulation on classification,labelling and packaging of substances and mixtures,amending andrepealing Directives67/548/EEC and 1999/45/EC,and amending Regulation(EC)No 1907/2006.

15.6.Chemical safety
assessment no data**16.OTHER INFORMATION**

The review affected this chapters:

1.3,3.3.3,3.3.4,4.2,4.5,4.6,5.5,5.6,6.4,7.2.2,7.2.3,7.3,8.1,8.2.1.1,10.2,10.5,10.6,10.7
11.1.1,11.1.2,11.1.3,11.2.3,11.2.2,11.2.1,11.3,11.4,11.5.1,11.5.2,11.5.3,12.1.1,12.1.2
12.1.3,12.2,12.3.1,12.3.2,12.3.3,12.4,12.6,12.5,14,15.14,15.11,15.10,16.3

Risk phrases

R8-Contact with combustible material may cause fire.
R40-Limited evidence of a carcinogenic effect.

H-phrases

H270-May cause or intensify fire;oxidiser.
H280-Contains gas under pressure;may explode if heated.

16.1.General information

In case of products and mixtures,it should be ensured that no new risks arise.
The information on this data sheet is based on our best knowledge at the time of printing the safety data sheet and is provided in good faith.However,certain data are being reviewed.
Users should note the potential for additional risks in case of using the product for purposes other than the recommended application.This data sheet may be used and reproduced for prevention and safety purposes only
The references to legislation,regulations and practical rules,and documents should not be considered complete.
It is the responsibility of the person receiving the product to consult all documents related to the use and handling of the product
The responsibility of parties handling the product also includes to pass on the whole of the information listed on the safety data sheet and necessary for work safety and for the protection of health and the environment,to the next person who may get in contact in any way with the product(use,storage,cleaing of containers,other operations).

16.2.Technical advisory services

Guangdong Huate Gas.,Ltd.
Jinfeng,Road,Heshun Lishui
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Tel:757-85123408
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Date: 2017-5-17	Nitrous oxide supplied in cartridges	Site: 9/9
Date of Revision: 2024-11-14		Version: 7
16.3. Sources		

Safety data sheet-United Brand.(2005-5-17)

16.4. Further information

